Document Control

Title	
Neutropenic Sepsis in Adults (Including management of Febrile Neutropenia / Neutropenic fever) Guideline	
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Directorate Diagnostics	**Department** Microbiology

Version	Date Issued	Status	Comment / Changes / Approval
0.1	Nov 2010	Draft	Draft after consultation with haematology and microbiology consultants
0.2	Nov 2010	Draft	Comments from Charlotte Overney. Associated documents and care in side room added.
0.3	Nov 2010	Draft	Contact details added. Imipenem changed to meropenem.
0.4	Nov 2010	Draft	Change of format
0.5	Jan 2011	Draft	Minor amendments to formatting and document control report. Hyperlinks to appendices.
0.6	Jan 2011	Draft	Minor amendments to title and formatting by TL.
0.7	Jan 2011	Draft	Change to HRCT referral criteria after input from Alison Moody
1.0	Jan 2011	Final	Approved by Drugs and Therapeutics Group in January 2011 – with amendments to tazocin, addition of emphasis to penicillin allergy and removal of 'more than' signs.
1.1	Jan 2012	Revision	Amendment to definition of neutropenic sepsis (appendix A). Update to header and footer and document control report by Corporate Governance.
1.2	Jan 2013	Revision	Addition of lactate to screening blood tests. CXR not routinely recommended. Stop antibiotics once responded to treatment irrespective of neutrophil count.
1.3	Feb 2013	Revision	Change to CXR wording after discussion in AWG. Addition of acute hospital setting to introduction and purpose. Change to meropenem doing to QDS. Addition of pneumococcal urinary antigen testing.
1.4	Jun 2013	Revision	Minor amendments by Corporate Governance to document control report and formatting.
2.0	Dec 14	Final	Review by Consultant Microbiologist, no amends. Published on

			Bob.
2.1	July 2016	Revision	Document inserted into new trust guideline template, updated references, phrase regarding time since chemotherapy elapsed removed as per NICE guideline. Rejected by DTC.
2.2	Aug 2016	Revision	Dosing regime for piperacillin-tazobactam altered to every 6 hours. Addition of ED poster, timing since chemotherapy altered to 6 weeks, addition of comment regarding *Aspergillosis* in further management. Rejected by DTC
2.3	Nov 2016	Revision	Amendments to Appendix 2 layout and insertion of timescale for review of antibiotics in febrile patients. Adjustment to Appendix 3 ED poster with phrase about blood cultures from pre-existing lines.
2.4	August 2018	Revision	Amendments to phrase on ED poster re: unwell and recent chemotherapy. Updated Trust logo on document header.
3.0	Nov 2018	Revision	Restructured appendices to merge with RDE document for shared guideline agenda. Insertion of MASCCC score and early discharge assessment criteria. Deletion of specimen sample pots. Update to appendix 5 ED poster. Approved at DTC Jan 2019.
3.1	Oct 2019	Revision	Updated approval process. Addition of warning re: methotrexate interaction with penicillin. Addition of warning card appendix 6. Approved by IPDG 28th Jan 2020

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Lead Director
Director of Medicine

Superseded Documents
Neutropenic Sepsis Guidelines
(Including management of Febrile Neutropenia / Neutropenic fever) v3.0

Issue Date	**Review Date**	**Review Cycle**
Jan 2020	Dec 2022	Three years

Consulted with the following stakeholders:

- Antibiotic Working Group
- Haematology
- Acute Oncology

Approval and Review Process

- Clinical Audit and Guidelines Group

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1. Purpose

1.1. This document sets out Northern Devon Healthcare NHS Trust's best practice guidelines for appropriate microbiological investigation and antimicrobial prescribing in adult patients with neutropenic sepsis.

1.2. This guideline applies to all adults and must be adhered to. Special considerations exist for pregnant and breastfeeding patients; liaise with specialist clinicians as appropriate in these cases. See separate guidance for paediatric patients.

1.3. Non-compliance with this guideline may be for valid clinical reasons only. The reason(s) for non-compliance must be documented clearly in the patient's notes.

1.4. This guideline is primarily aimed at all prescribing teams but other staff (e.g. nursing staff, pharmacists) may need to familiarise themselves with some aspects of the guideline.

1.5. Implementation of this guideline will ensure that:

- Patients with neutropenic sepsis are managed according to current evidence and standards of practice in the wider healthcare community.
- A standard of care is specified to facilitate a consistent approach between haematology/oncology, microbiology and pharmacy in terms of patient management, specimen processing and drug availability.

2. Definitions

Febrile Neutropenia

2.1. Febrile neutropenia is defined as fever of 38°C or above, in the presence of neutropenia, where:

- Neutrophil count is less than 0.5 x 10^9/L, **or**
- Neutrophil count is less than 1.0 x 10^9/L and predicted to fall below 0.5 x 10^9/L)

Neutropenic Sepsis

2.2. Neutropenic sepsis is defined as haemodynamic instability in the presence of neutropenia (as defined above) and possible infection. The patient may not be febrile.

3. Responsibilities

3.1. Responsibility for education and training lies with the Lead Consultant Microbiologist for Antibiotic Stewardship. It will be provided through formal study days and informal training on the ward.

3.2. The author will be responsible for ensuring the guidelines are reviewed and revisions approved by the Clinical Effectiveness Committee in accordance with the Document Control Report.

3.3. All versions of these guidelines will be archived in electronic format by the author within the Antibiotic Stewardship policy archive.

3.4. Any revisions to the final document will be recorded on the Document Control Report.

3.5. To obtain a copy of the archived guidelines, contact should be made with the author.

3.6. Monitoring of implementation, effectiveness and compliance with these guidelines will be the responsibility of the Lead Clinician for Antibiotic Stewardship. Where non-compliance is found, the reasons for this must have been documented in the patient's medical notes.

Role of Antibiotic Working Group (AWG)

3.7. The AWG is responsible for:

- Leading antibiotic guideline development and review within Northern Devon Healthcare Trust
- Involving all relevant stakeholders in guideline development and review

4. Contacts

4.1. Contact numbers:

- Microbiologist Bleep 193. Via switchboard out of hours.
- Acute Oncology Service Bleep 240 (Nurse)
- Antibiotic Pharmacist Bleep 029 (Mon-Fri only)
- Haematology Junior Bleep 208 (Mon-Fri only)
- Haematology CNS Bleep 203 (Mon-Fri only) (out of hours, leave a message on the answerphone ext 3198)
- Oncologist Seamoor Unit (Mon-Fri). Via switchboard out of hours

5. Investigations

5.1. Samples for pathology

- Send clotted blood for U+Es, LFTs (including albumin), CRP
- Send EDTA blood for FBC
- Send fluoride blood (grey top) for lactate (or use ABG)

5.2. Samples for microbiology

- Take blood cultures (peripheral and central lines) and urine culture
- If a chest focus is suspected then send:
  - Urine (white top universal) for Legionella and Pneumococcal antigen detection.
  - Throat swab (viral transport medium) for virus PCR
- If patient has diarrhoea send a stool sample for *C. difficile* toxin testing

5.3. Monitoring

- Blood pressure
- Temperature
- Pulse
- Vascular device inspection (if present)
- Fluid balance
- Airway / oxygen saturations
- Behaviour / mental state
- Skin, and soft tissues at extremities (inspect for rashes, capillary refill time)

5.4. Imaging

Discuss with radiologist. Consider:

- Chest X-ray. Should normally be taken, but may be omitted if no clinical suggestion of pneumonia or lung abscess.

6. Management of Febrile Neutropenia and Neutropenic Sepsis

6.1. See appendix 1

7. Monitoring Compliance with and the Effectiveness of the Guideline

Standards/ Key Performance Indicators

7.1. Key performance indicators comprise:

- Percentage of patients with blood cultures taken on admission
- Percentage of patients receiving antibiotics within one hour of arrival

8. References

- Hughes WT et al. 2002. *Guidelines for the Use of Antimicrobial Agents in Neutropenic Patients with Cancer*. Clin Infect Dis 2002; 34: 730-751
- Rhodes et al. 2016. *Surviving sepsis campaign : International guidelines for management of severe sepsis and septic shock 2016.* https://journals.lww.com/ccmjournal/Fulltext/2017/03000/Surviving_Sepsis_Campaign___International.15.aspx
- NICE. 2016. *Sepsis: recognition, diagnosis and early management.* https://www.nice.org.uk/guidance/ng51/resources/sepsis-recognition-diagnosis-and-early-management-1837508256709
- NICE and the National Collaborating Centre for Cancer (NCCC). 2012. Neutropenic sepsis: prevention and management of neutropenic sepsis in cancer patients. Evidence review, search strategies, health economics evidence review and health economics plan. https://www.nice.org.uk/guidance/cg151/evidence/evidence-review-pdf-188303582

9. Associated Documentation

- Penicillin Allergy Policy
- Incident Reporting Policy
- Antibiotic Prescribing Policy
- Management of sepsis and severe sepsis policy

Appendix 1: Management of Neutropenic Sepsis and Febrile Neutropenia

10.1. Name of guideline on app

Neutropenic Sepsis

10.2. Location on app

Secondary Care >
Adult treatment >
Severe Sepsis and Septic Shock >
Neutropenic Sepsis

10.3. Header

- Use this protocol if the patient has a:
  - Neutrophil count below 0.5 x 10^9/L OR neutrophil count awaited and chemotherapy within the past six weeks, OR neutrophil count below 1.0 x 10^9 /L and expected to fall, OR bone marrow transplant patient.

  AND

  - Temperature above 38° or below 35°C OR history of documented fever/rigors OR systemically unwell OR unexplained abnormal systolic blood pressure

10.4. Initial Investigations for all patients

- BEFORE starting antibiotics:
  - Take cultures (blood and urine). If central and peripheral lines in situ, take simultaneous blood cultures from both (10mL per bottle optimises yield).
- As soon as possible:
  - MRSA screen – nose, throat and wound swabs (if previous MRSA positive add perineal swab, line swab and urinary catheter sample)
  - Chest X-ray (within 4 hours if chest focus)
  - Additional microbiology samples, depending on suspected focus
    - Chest: sputum for MC&S, viral throat swab for respiratory virus PCR, atypical pneumonia serology (clotted blood tube – include illness onset details). *If CURB≥3, send urine for pneumococcal and legionella antigens*
    - GI: stool sample for MC&S, C.difficile
    - Oropharyngeal: viral swab for perioral Herpes, swab for fungal culture if previous treatment failure with antifungal for oropharyngeal thrush
    - Line infection: Exit site swab for MC&S

10.5. Treatment. [closed]

- Before giving first dose of antibiotics: check the sensitivities of recent isolates on the pathology system. Also check for patient alerts on MRSA, ESBLs or any other infection control flags on the admissions computer system.
  - If resistant isolates not covered by protocol antibiotics discuss with microbiology
- Be aware: Interaction between methotrexate and penicillins causing reduced excretion of methotrexate and potential for bone marrow toxicity. Follow mild penicillin allergy treatment protocol for patients on methotrexate.

First-Line

Piperacillin-Tazobactam (Tazocin) 4.5g QDS IV via infusion over 30 minutes

If evidence of severe sepsis, then also give a single dose of

Gentamicin IV as per Trust protocol

Do not give further gentamicin doses without advice. If patient remains febrile, discuss with microbiology / haematology / oncology

Mild Penicillin allergy:

Meropenem 500mg QDS IV

If evidence of severe sepsis, then also give a single dose of

Gentamicin IV as per Trust protocol

Do not give further gentamicin doses without advice. If patient remains febrile, discuss with microbiology / haematology / oncology

Severe Penicillin allergy:

Aztreonam 2g TDS IV

Plus

Vancomycin IV as per Trust protocol

Suspected line infection:

Consider adding Vancomycin IV as per Trust protocol

Suspected chest focus:

Add Clarithromycin* 500mg BD PO

Also consider influenza – particularly in flu season

**NB. Caution: multiple interactions with CYP450-metabolised drugs when prescribing clarithromycin*

Suspected abdominal focus (with possible *C. difficile* infection):

Add Metronidazole 400mg TDS PO

Oropharyngeal infection

Candida:

Nystatin suspension 1mL QDS,

or

Fluconazole 200mg OD PO (IV dose directly equivalent if nil by mouth)

Herpes simplex:

Aciclovir 400mg 5 times daily PO (for IV dosing, consult BNF – dosing regimen markedly different)

10.6. Red Flags [closed]

The progression of infection in neutropenic patients can be rapid, and neutropenic patients with early bacterial infections cannot be reliably distinguished from non-infected patients at presentation. Empirical antibiotic therapy should be administered promptly to all neutropenic patients at the onset of fever or evidence of possible sepsis.

10.7. Help notes [closed]

- Patients with febrile neutropenia / neutropenic sepsis should usually be nursed in a side room.

10.8. Neutropenic Sepsis Warning Cards [closed]

Neutropenic sepsis warning cards are held by qualifying patients who are managed by Northern Devon Healthcare Trust. The card is a “Patient Specific Direction”, similar to the inpatient drug chart.

The card will be signed by the Consultant Haematologist/Oncologist at the time the patient is consented for cytotoxic chemotherapy. The card will have a start date and a finish date specified, and cannot be used outside of that time.

The card permits a single dose of an appropriate antibiotic to be given according to the neutropenic sepsis policy, after appropriate investigations. The card is a single use document, and should be countersigned by the Nurse administering the dose of antibiotics, and then filed in the patient’s notes on permanent record.

The Consultant Haematologist/Oncologist reviewing the patient during their admission will need to write a new card if they still wish the patient to be able to access this treatment protocol.

Cards from outside the Northern Devon Healthcare Trust area should not be used to administer antibiotics against, as they are not a valid prescription.

10.9. Organism and sensitivity data [closed]

Organisms and sensitivity data (2008-10)		Taz	Gent	Cip	Mer	Ceftaz
Common	Pseudomonas aeruginosa	100%	95%	100%	90%	95%

10.10. Version control [closed]

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Appendix 2: Suspected Febrile Neutropenia Protocol for Adult Haematology and Oncology: Initial Assessment

USE THIS PROTOCOL IF PATIENT HAS: Neutropenia i.e. neutrophils <1.0×10^9/L **or** suspected neutropenia (chemotherapy in last 6 weeks)

AND EITHER: a) History of fevers / rigors (temperature >38 °C or <35°C)

OR: b) Clinical signs of sepsis (HR >90bpm, RR >20/min, SBP <90mmHg) or obvious signs of infection

↓

Initial Investigations for All Patients

1. **BEFORE starting antibiotics:** **Blood cultures** (peripheral and central/PICC line at the same time, 10mL per bottle optimises yield)
   Urine for culture
2. **As soon as possible:** **MRSA screen** nose, throat, wound swabs. If previous MRSA positive, add perineum, lines and urinary catheter
   Additional microbiology samples dependent on focus
   Blood tests to include: FBC and clotting, renal + U&Es, LFTs, bone, CRP, lactate, glucose, magnesium

↓

Check admissions computer system for patient alerts: MRSA, ESBLs other infection control flags.
Check pathology system for recent isolates and sensitivities to intended empirical antibiotics Tazocin®, meropenem, vancomycin, aztreonam

If not covered by protocol antibiotics below, discuss with microbiology

↓

START EMPIRICAL ANTIBIOTICS WITHIN 1 HOUR OF ARRIVAL

First dose must be given within one hour of ward arrival, or temperature spike if current inpatient. Use PGD/PSD where available to avoid delay

FIRST LINE	SECOND LINE (mild penicillin allergy)	THIRD LINE (severe penicillin allergy)
Piperacillin-tazobactam (Tazocin®) 4.5g QDS IV	Meropenem 500mg QDS IV	Aztreonam 2g TDS IV + Vancomycin IV as per Trust protocol

↓

WHEN THERE ARE FOCAL SIGNS OF INFECTION add targeted antimicrobials and take specific samples

↓

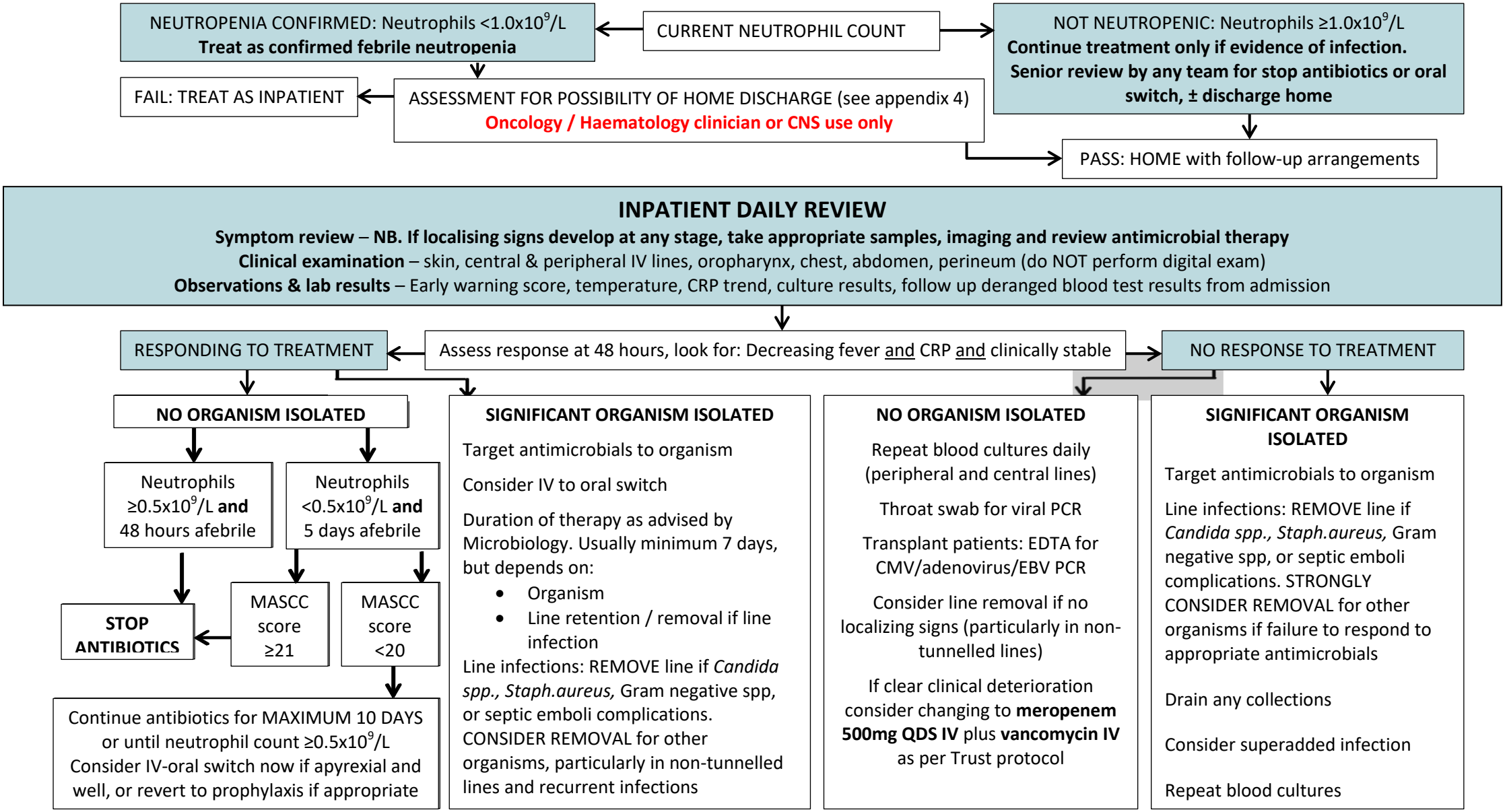
CHEST: Add **clarithromycin 500mg BD PO/IV,** consider influenza in flu season
Samples: MC&S sputum, respiratory virus PCR throat swab, clotted blood atypical pneumonia serology, *if CURB≥3, urine pneumococcal and legionella antigens*
Investigations: Chest X-ray within 4 hours of arrival

GI: TYPHILITIS / PERIANAL SEPSIS (diarrhoea / pain):
Add **metronidazole 400mg TDS PO**
Samples: Faeces pathogen culture, C.difficile toxin
Investigations: Do not perform DRE

OROPHARYNGEAL: *CANDIDA*:
Add **nystatin suspension 1mL QDS PO *or* fluconazole 200mg OD PO/IV**
Samples: Fungal mouth swab if previous treatment failure
OROPHARYNGEAL: PERIORAL HERPES SIMPLEX:
Add **aciclovir 400mg 5 times daily (see BNF for IV)**
Samples: viral swabs

CENTRAL LINE INFECTION: Add **vancomycin IV as per Trust protocol**
Samples: Blood cultures (central and peripheral line simultaneously), exit site swab
Actions: Remove line if tunnel infection, or no response to antibiotics

Appendix 2: Ongoing Management of Neutropenic Sepsis and Febrile Neutropenia: Ongoing management 24 - 48 hours



Appendix 3: Ongoing Management of Neutropenic Sepsis and Febrile Neutropenia beyond 48 hours

INPATIENT DAILY REVIEW

Symptom review – **NB. If localising signs develop at any stage, take appropriate samples, imaging and review antimicrobial therapy**
Clinical examination – skin, central & peripheral IV lines, oropharynx, chest, abdomen, perineum (do NOT perform digital exam)
Observations & lab results – Early warning score, temperature, CRP trend, culture results, follow up deranged blood test results from admission

AFEBRILE, WITH DECREASING CRP AND CLINICALLY STABLE AT 96 HOURS

NO ORGANISM ISOLATED

SIGNIFICANT ORGANISM ISOLATED

FOLLOW GUIDANCE IN APPENDIX 2:

ONGOING MANAGEMENT OF NEUTROPENIC SEPSIS AND FEBRILE NEUTROPENIC 24-48 HOURS

If still febrile after 4 days of antibiotics (sooner if clinical suspicion of invasive *Aspergillosis*), CRP persistently raised or patient clinically unwell

Repeat blood cultures from central and peripheral lines at the same time. **Request urgent HRCT of chest & upper abdomen**

CLINICAL ± HRCT SUSPICIOUS OF INVASIVE FUNGAL INFECTION

Arrange BAL if patient fit enough

Consider CT guided biopsy

Add antifungal Rx (Caspofungin unless contra-indicated)

NO CLINICAL OR HRCT EVIDENCE OF INFECTION

PATIENT STABLE & NOT SEVERELY UNWELL

Discuss with Microbiology

Continuing existing treatment may be the best decision

Systemic antifungal probably not indicated

CT NO EVIDENCE OF INFECTION

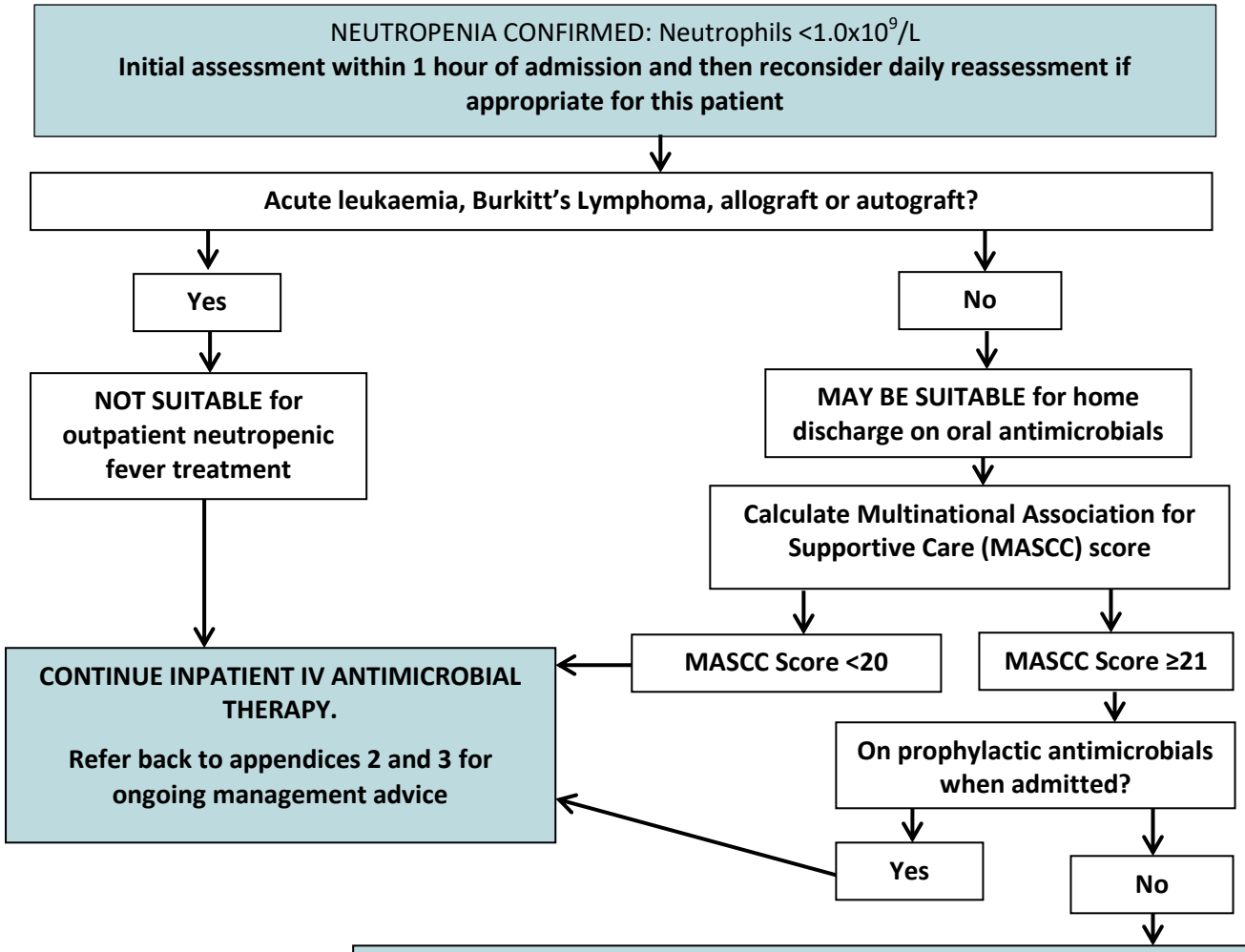
PATIENT DETERIORATING OR SERIOUSLY UNWELL

Discuss with Microbiology
Consider: - Additional imaging: abdominal USS, CT sinuses
- BAL/NPA respiratory viruses, if patient fit enough
- *Pneumocystis jiroveci*

Add antifungal treatment if high risk for invasive fungal infection (persistent neutropenic fever >4-7 days and expected neutropenia >7 days)

Antifungal choice depends on whether received antifungal prophylaxis or not. If unsure, prescribe **Caspofungin unless contra-indicated**

Appendix 4 – HAEMATOLOGY / ONCOLOGY SPECIALISTS and CNS use only: Assessment for Early Discharge of Low-Risk Neutropenic Patients



Multinational association for supportive care (MASCC) score		
Characteristic		Score
Age	≥60	0
	<60	2
Patient dehydrated, requiring fluids?	No	3
	Yes	0
COPD?	Yes	0
	No	4
Blood pressure (hypotensive)	Systolic BP <90mmHg	0
	Systolic BP ≥90mmHg	5
Solid tumour?	Yes	4
	No	0
Was the patient already an inpatient?	Yes	0
	No	3
Symptoms related to febrile episode?	None/ mild	5
	Moderate	3
	Severe	0

Change to oral antibiotics and treat as inpatient for at least 24 hours.
Repeat FBC, CRP and clinical assessment next day: If well and responding to treatment consider early discharge on oral antibiotics.
Target antimicrobial therapy according to culture results, if positive.
Empirical treatment: 1st line – co-amoxiclav 625mg TDS PO 7 days OR penicillin allergy (mild or severe) – ciprofloxacin 500mg BD PO 7 days

Senior review by consultant or specialist trainee after 24 hours. Discharge if clinically stable, temperature <37.5°C, CRP improving, and assured of good patient compliance & social circumstances for readmission. Check follow-up appointment booked and advise contact Acute Oncology Service / haematology urgently if further pyrexia or unwell.

Appendix 4 contd. – HAEMATOLOGY / ONCOLOGY SPECIALISTS and CNS use only: Assessment for Early Discharge of Low-Risk Neutropenic Patients

Low risk febrile neutropenia policy: outpatient management protocol

- Background
  Traditionally, all patients with suspected neutropenic sepsis have been admitted and managed with broad spectrum intravenous antibiotics. Increasingly, it has been recognised that there is a subset of patients where the risk of complications related to neutropenic sepsis or febrile neutropenia is low.

  The MASCC score was developed in 2000 as a tool for identifying low risk patients. NICE has suggested that this may be an appropriate tool to identify patients, for whom treatment can safely be de-escalated to facilitate early discharge, when utilised by specialist practitioners in haematology/oncology (NICE and NCCC, 2012).

- Procedure
  Initially, any patient attending the outpatient cancer services units (Cherrybrook Daycase Unit or the weekend triage unit at RDE; or Seamoor Unit at NDDH) with suspected neutropenic sepsis should receive a stat dose of intravenous antibiotics within 1 hour of arrival – as per the existing protocol.

  A MASCC score should then be calculated and the patient categorised as high or low risk. Patients in the high risk group (MASCC <20) should be admitted as per existing guidelines, to continue broad spectrum intravenous antibiotics. Those patients with MASCC ≥21 can be considered for early discharge with oral broad spectrum antibiotics.

- Other Considerations
  Does the patient live alone?
  - If the patient deteriorates, they need to be able to return to hospital for admission or intravenous antibiotics. Those living alone may be deemed unsafe for outpatient management for this reason.

  Can the patient readily return for clinical review?
  - The patient must be able to attend the outpatient cancer services unit the following day – telephone triage would not be appropriate.

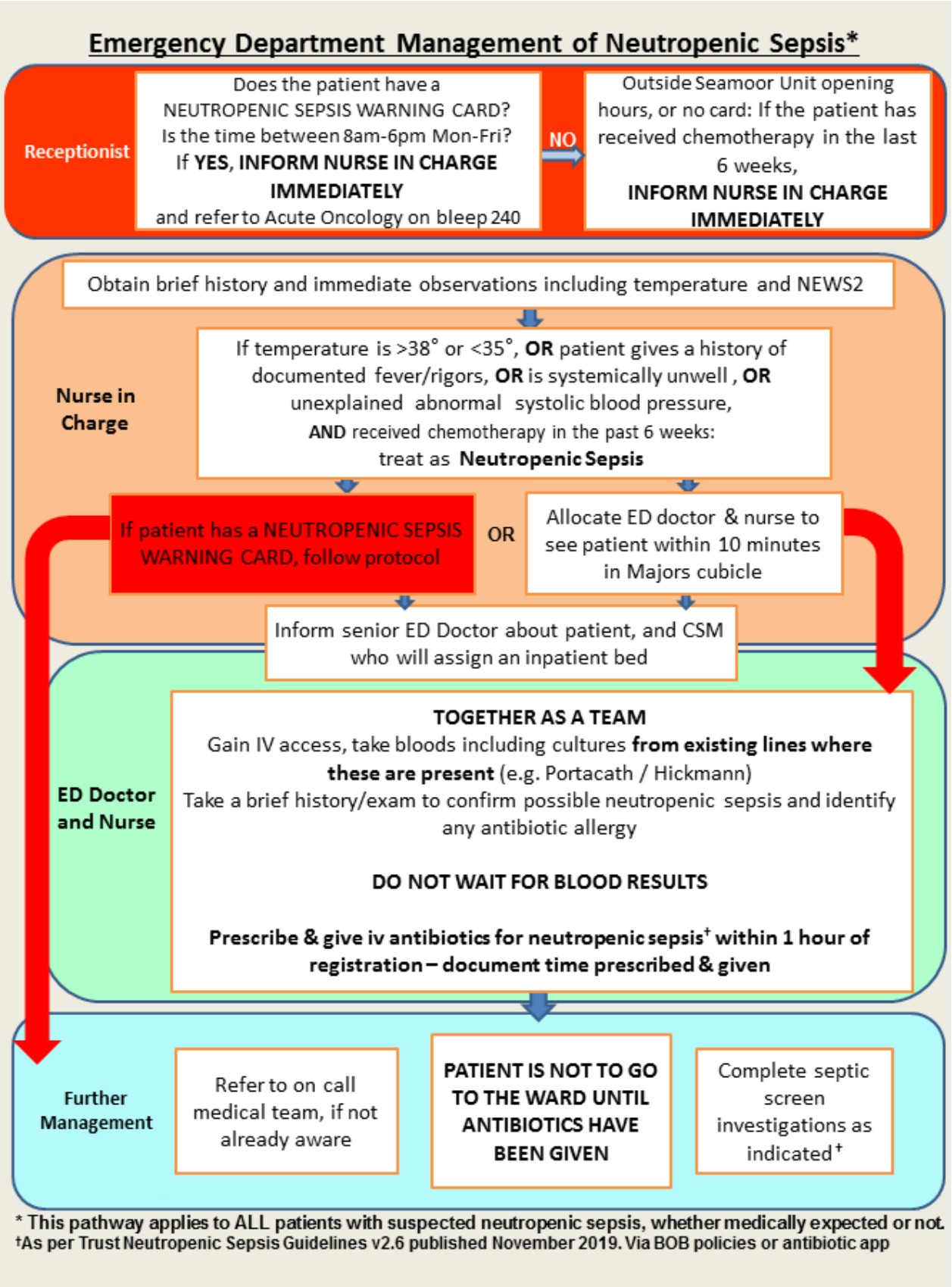
  Does the patient have any evidence of organ compromise?
  - Renal or hepatic failure may affect the absorption/effectiveness/excretion of antibiotics and therefore the patients should be managed in a setting where this can be dealt with appropriately.

  Does the patient have any co-morbidities?
  - The elderly or those patients with potentially complicating co-morbidities may require a more hospital-based focus of care in case of sudden deterioration.

  Is there an obvious infection that may require prolonged antibiotic management?
  - Some specific infections, such as cellulitis, may require prolonged use of IV or combination antibiotics. These patients should initially be managed as inpatients. Once stable, plans for oral switch or outpatient parenteral antibiotic therapy (OPAT) should be made.

Appendix 5: Neutropenic Sepsis Flowchart for Emergency Department

Image can be accessed for printing via BOB > Departments > ED > Policies and Guidelines (powerpoint file)



Appendix 6: Neutropenic Sepsis Alert Card

10.11. **Not to be printed –** cards are available from Seamoor Unit to be filled out during the appointment as part of the consent to neutropenia-causing treatment. The patient should leave the appointment with the card filled in and have been counselled on its use.

10.12. The dose of vancomycin for severely penicillin allergic patients must be specified on the card

10.13. The card should have a start and end date specified

10.14. Patients who are admitted for neutropenic sepsis treatment must be provided with a replacement card before leaving hospital, if they will be continuing with neutropenia-causing treatment in the future.

Doctor/NMP: delete unwanted option by crossing out the box.
Sign and date desired antibiotic therapy to be used in case of emergency.

First line treatment:
Piperacillin-tazobactam (Tazocin ®) 4.5g intravenous infusion over 30 minutes
(reconstitute as per manufacturer's directions)

Doctor/NMP: ____
Date: ____ / ____ / ____
GMC/NMC number: ____
Administered by: ____
NMC no: ____
Date: ____ / ____ / ____ Time: ____

Mild Penicillin Allergy:
Meropenem 500mg intravenous bolus by slow injection over 5 minutes
(reconstitute as per manufacturer's directions)

Doctor/NMP: ____
Date: ____ / ____ / ____
GMC/NMC number: ____
Administered by: ____
NMC no: ____
Date: ____ / ____ / ____ Time: ____

Severe Penicillin Allergy:
Aztreonam 2g intravenous bolus by slow injection over 5 minutes, PLUS
Vancomycin intravenous infusion as per dose specified

Doctor/NMP: ____
Date: ____ / ____ / ____
GMC/NMC number: ____
Aztreonam administered by: ____
Date: ____ / ____ / ____ Time: ____
Vancomycin dose: ____
Vancomycin administered by: ____
NMC no: ____
Date: ____ / ____ / ____ Time: ____

Then continue treatment as per neutropenic sepsis guideline on Trust intranet

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NHS

Neutropenic Sepsis Alert Card

For use in Northern Devon Healthcare NHS Trust
Emergency Department, Minor Injury Units or Seamoor Unit

Patient addressograph label here

Patient Allergies:

Not to be printed – example only

This patient is receiving drugs that suppress the immune system. They are at risk of developing neutropenic sepsis.

They have been consented to receive appropriate urgent care using this neutropenic sepsis warning card.

Valid from: ____ / ____ / ____

Valid until: ____ / ____ / ____

This patient-specific direction provides authority for Registered Nurses qualified in administration of intravenous drugs to administer a single dose of antibiotics in an Accident an Emergency, Minor Injuries Unit or Seamoor Unit based within North Devon Healthcare NHS Trust.

Use this protocol if the patient has a:

Neutrophil count below 0.5 x 10^9/L
OR neutrophil count awaited and chemotherapy in last 6 weeks **OR** bone marrow transplant

AND

Temperature above 38°C or below 35°**OR** history of documented fever/rigors **OR** systemically unwell.

If the patient presents with signs of severe sepsis, blood pressure <90mmHg or difficulty breathing, immediately refer to duty doctor before continuing.

Immediately on arrival:

1. Take blood cultures (including from peripheral and central lines if present) and urine culture.
2. Send clotted blood for U+Es, LFTs, CRP
3. Send EDTA blood for FBC
4. Send fluoride blood (grey top) for lactate, or use ABG machine if available.

As soon as possible:

5. Additional microbiology samples, depending on suspected focus:
   a. MRSA screen – nose, throat and wound swabs (if previous MRSA positive add perineal swab, line swab and urinary catheter sample)
   b. Chest X-ray (within 4 hours if febrile)
   c. Chest: sputum for MC&S, viral throat swab for respiratory virus PCR, atypical pneumonia serology (clotted blood tube – include illness onset details). If CURB≥, send urine for pneumococcal and legionella antigens
   d. GI: stool sample for MC&S, C.difficile
   e. Oropharyngeal: viral swab for perioral Herpes, swab for fungal culture if previous treatment failure with antifungal for oropharyngeal thrush
   f. Line infection: Exit site swab for MC&S

Ensure all sample containers are adequately labelled with patient details

6. Check for sensitivities of recent isolates on Pathology system. If not covered by protocol antibiotics, this patient should be seen by a duty Doctor immediately.
7. If patient is still suitable for treatment using this patient-specific direction, administer a stat dose of antibiotics within 60 minutes of arrival, as overleaf:

Not to be printed – example only